Alen Issac Sam

bt23btech11003@iith.ac.in | alenissacsam05@gmail.com | +91 9645873057 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.59
Aravinda Vidyamandiram School	Class XII (CBSE)	2022	91.80%
MGM Public School	Class X (CBSE)	2020	91.20%

EXPERIENCE

Machine Learning Industrial Trainee - Intrainz

- Completed a certified industrial ML training programme; recognised by the Head of Operations for performance and initiative.
- Expertise gained: supervised learning pipelines end to end -- preprocessing, feature engineering, model evaluation.

Independent Blockchain & Quantitative Systems Developer - Solo-owned production projects

- Sole architect and engineer on shipped systems across smart contracts, clinical SaaS and quantitative research, several delivered inside 48-hour hackathon sprints with cross-functional teams.
- Expertise gained: contract security review, gas optimisation, test coverage discipline (>95% unit coverage on the stablecoin suite), and honest measurement of whether a system's edge is real.

PROJECTS

Dislocation_bps - Measurement Instrument for Solana AMM Arbitrage

- Built a live mainnet instrument that prices every visible arbitrage cycle and records what *would* have cleared -- 26,131 opportunities over 15 hours, 669 clearing, $2.50 realised, and the finding that the median opportunity is worth $0.0013 against a 2 bps cheapest round trip.
- Proved with data (not argument) that opportunity size and lifetime run in opposite directions -- every gap worth over $0.10 was gone before the next slot -- answering "add capital" and "go faster" quantitatively; a flash loan would have added nothing.
- Live execution built behind four safety switches and off by default, because the instrument's own measurements say not to trade it.
- **Tech:** Rust, Solana, Tauri, SQLite, CI/CD

ClinicFlow - Clinic & Patient Management for Solo Doctors

- Owned a complete product: appointments with live queue, EMR-lite visits, PDF prescriptions, FEFO batch-tracked pharmacy, insurance/TPA workflow and billing -- with allergy and drug-drug interaction screening that advises, never blocks.
- Shipped Indian healthcare compliance: DPDP export/delete, Verhoeff-checked ABHA identity, FHIR R4 OPConsultRecord export, ABDM consent artefacts, plus WhatsApp Cloud API reminders and offline-first consults.
- **Tech:** Next.js 16, TypeScript, Supabase (Postgres + RLS), Tailwind v4, Vercel Cron, react-pdf, Vitest

StablePitch - Parametric Prediction Markets on Solana ·

- Designed fixed-payout binary options on live match statistics, priced by an off-chain Poisson goal model that delivers ed25519-signed quotes against a USDC House Vault.
- Removed house insolvency risk by reserving collateral per position at mint; settlement CPIs into TxLINE's on-chain validateStat oracle, verified against per-minute Merkle roots, with a receipt UI that renders the full proof chain.
- **Tech:** Rust/Anchor, Solana, Python pricer, React, TxLINE SSE feeds, Merkle proofs

Brand & Buzzword Hangman

- Ensemble of miss-conditioned dual-head Transformers playing Hangman on unseen words: **70.94% win rate** over 250,000 test words (69.01% single model), each model trained in ~100 minutes on one 6 GB laptop GPU.
- Reverse-engineered the grader's resolution of an ambiguity in the rules by matching the leaderboard score to eight decimal places from a strict local simulation.

- **Tech:** PyTorch, Transformers, ensembling, reproducible end-to-end notebook

Intelligent Cloud Architect & Geospatial Analysis

- Google ADK multi-agent system turning plain-English infrastructure requirements into security-validated AWS architectures with generated Terraform (Kaggle Agents Intensive capstone).
- Ernakulam shoreline change 1995-2024: Google Earth Engine extraction feeding a custom Python DSAS emulator -- tidal correction, transect casting, NSM/EPR and RMSE.
- **Tech:** Python, Google ADK, Gemini 2.0, Terraform, Google Earth Engine, QGIS, GeoPandas

TicketVerse - On-Chain NFT Ticketing

- Built an anti-counterfeit, royalty-aware ticketing platform in a 48-hour sprint, later extended into a Next.js marketplace with wallet-gated admin access and IPFS metadata export.
- **Tech:** Solidity, Avalanche, Next.js 16, Prisma/PostgreSQL, Wagmi/Viem, Three.js/R3F, GSAP

SKILLS

Technical Skills

- **Blockchain & Web3:** Solidity, Foundry, Rust/Anchor, PyTeal, Ethereum, Solana, Algorand, Avalanche, Chainlink CCIP, ERC-20/721/1155, smart contract security, gas optimisation, DeFi & AMM microstructure, IPFS, zero-knowledge proofs, decentralized identity (DID), ethers.js, Wagmi, Viem
- **Quantitative & Trading:** market microstructure, arbitrage measurement, Hidden Markov Models, time-lagged cross-correlation, Dynamic Time Warping, cointegration, Poisson derivatives pricing, NinjaScript strategy development
- **Machine Learning and AI:** PyTorch, TensorFlow, scikit-learn, XGBoost, Transformers, CTC/beam search decoding, time-series forecasting, feature engineering, NumPy, Pandas, Google ADK
- **Languages:** Solidity, Rust, Python, TypeScript, JavaScript, C++, C, SQL, HTML/CSS
- **Web & Backend:** Next.js, React, Node.js, Tailwind CSS, React Query, Prisma, Supabase, PostgreSQL, REST APIs, WebSockets/SSE, Vercel
- **Geospatial & Tools:** Google Earth Engine, QGIS, Git & GitHub, Docker, GCP, Linux, Jupyter
- **Spoken:** English, Hindi, Malayalam (native), Japanese (limited working)

Leadership & Soft Skills

- **Leadership & Ownership:** Elected class representative for a full academic year; held three concurrent coordinator portfolios across a six-month festival cycle; sole end-to-end owner of a client-facing clinical product.
- **Stakeholder Communication:** Coordinated between internal organizing teams and external sponsors; documents engineering trade-offs so non-technical readers can act on them.
- **Intellectual Honesty:** Published a negative result on own arbitrage system rather than overclaiming -- built the measurement that says not to trade it.
- **Collaboration & Adaptability:** Delivered across three 48-hour hackathons on unfamiliar chains (Solana, Algorand, Avalanche); picks up a new stack quickly.

RELEVANT COURSES

- **Computing & Algorithms:** Introduction to Programming · Artificial Intelligence · Sequence Alignment Algorithms · Big Data Biology and Biological Databases · Algorithms for Molecular Dynamics Simulation *(ongoing)*
- **Machine Learning & Data:** Machine Learning in Bioinformatics · R and Python for Biologists · Business Analytics using Python · Remote Sensing, GIS and Climate Change · AI in Drug Discovery *(ongoing)*
- **Mathematics & Statistics:** Probability and Random Variables · Biostatistics · Elementary Linear Algebra · Calculus I & II · Differential Equations
- **Finance & Economics:** Finance for Engineers · Game Theory for Startups · Introductory Economics · Introduction to Entrepreneurship

POSITIONS OF RESPONSIBILITY

Class Representative, IIT Hyderabad-- elected by the batch to channel student concerns into faculty and administrative forums.

Workshop, Infrastructure & Sponsorship Coordinator, ELAN & nvision, IIT Hyderabad -- three concurrent portfolios across a six-month festival cycle, coordinating internal teams and external partners.

ACHIEVEMENTS

Kaggle: Store Sales Time-Series Forecasting -- top 15% of 3,700 participants (RMSLE 0.58 0.40); Brain-to-Text Neural

Decoder -- 230 of 463 teams (WER 32.1 23.5 with CTC loss, beam search and 4-gram rescoring).

Hackathons: Avalanche National Hackathon (Draper House, Hyderabad); Algorand × T-Hub Decentralized Identity Hackathon; TxODDS Prediction Markets & Settlement, World Cup track -- Certificate of Appreciation, CodeKurukshetra.

Certifications: Cyfrin Updraft -- Solidity Smart Contract Development and Foundry Fundamentals (Jul 2025), Blockchain Basics (Jun 2025); 5-Day AI Agents Intensive with Google, Kaggle (Dec 2025); Machine Learning Industrial Training, Intrainz; Data & Analytical Skills, Crion Versity (May 2024).

NinjaTrader Holiday Futures Fest -- ranked 12 of 3,519 traders globally (top 0.3%) with an 88% win rate.

NinjaTrader Arena Cup Qualifier -- placed 232 of 10,000 in a global simulated futures competition with a $50,000 prize pool.

EXTRACURRICULAR

Chess (chess.com, active rapid and puzzles) and futures trading, both pursued as daily-practice hobbies.